11. BST Inorder Traversal

Problem Statement

Write Recursive Approach for Inorder Traversal of Binary Tree.

Given a binary tree, write a recursive algorithm to perform an inorder traversal of the tree and return the elements in the order they were visited.

Examples

Example 1:

```
Input:
           5
         /   \
        3     8
       / \   / \
      2   4 7   9

Output:
Inorder Traversal: [2, 3, 4, 5, 7, 8, 9]

Explanation:
The binary tree has the following structure:
           5
         /   \
        3     8
       / \   / \
      2   4 7   9

Performing an inorder traversal visits the nodes in ascending order: 2, 3, 4, 5, 7, 8,
9.
```

Example 2:

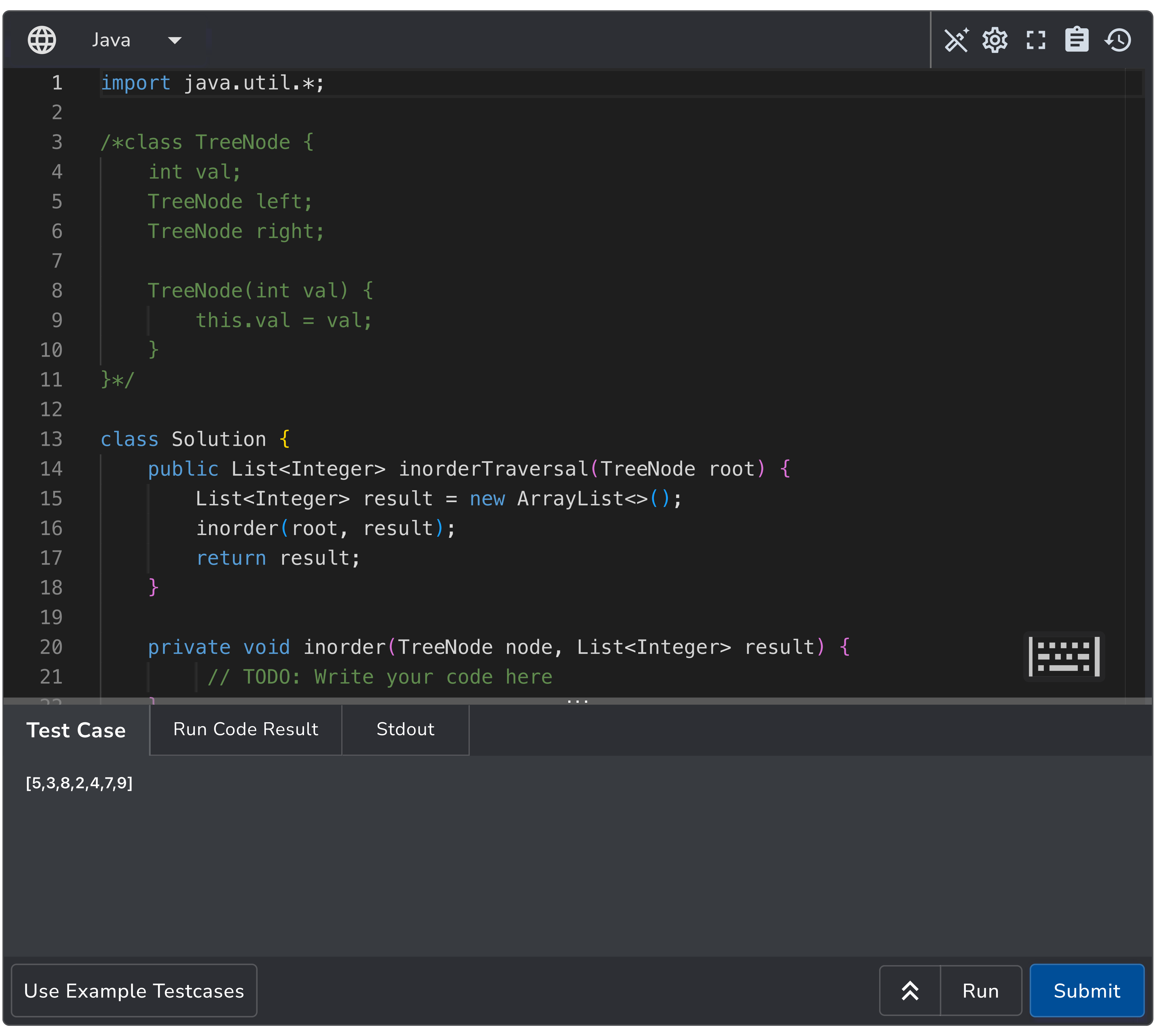
```
Input:
           10
         /    \
        6      15
       / \    / \
      3   8  12  18
            \
             9

Output:
Inorder Traversal: [3, 6, 8, 9, 10, 12, 15, 18]

Explanation:
The binary tree has the following structure:
           10
         /    \
        6      15
       / \    / \
      3   8  12  18
            \
             9

Performing an inorder traversal visits the nodes in ascending order: 3, 6, 8, 9, 10, 12,
15, 18.
```

Example 3:

```
Input:
           20
         /    \
        12     25
       / \    / \
      8   15  22  28
     / \       \
    6   10      24

Output:
Inorder Traversal: [6, 8, 10, 12, 15, 20, 22, 24, 25, 28]

Explanation:
The binary tree has the following structure:
           20
         /    \
        12     25
       / \    / \
      8   15  22  28
     / \       \
    6   10      24

Performing an inorder traversal visits the nodes in ascending order: 6, 8, 10, 12, 15,
20, 22, 24, 25, 28.
```

Try it yourself

Try solving this question here:



```java
import java.util.*;

/*class TreeNode {
    int val;
    TreeNode left;
    TreeNode right;

    TreeNode(int val) {
        this.val = val;
    }
}*/

class Solution {
    public List<Integer> inorderTraversal(TreeNode root) {
        List<Integer> result = new ArrayList<>();
        inorder(root, result);
        return result;
    }

    private void inorder(TreeNode node, List<Integer> result) {
        // TODO: Write your code here
```

[5,3,8,2,4,7,9]

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Solution: Inserting A New Node In A BST

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Solution: BST Inorder Traversal

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